

Science question paper

SUBJECT

Science

CLASS

10

TOPIC

Electricity

MAXIMUM MARKS

50

TIME

1 Hour

General Instructions:

1. The question paper consists of five sections A, B, C, D and E.
2. All questions are compulsory.
3. Section A consists of multiple choice questions.
4. Section E consists of case-study based questions.
5. Diagrams should be neat and properly labelled wherever required.

Section A - Multiple Choice Questions

Choose the correct option.

10 × 1 = 10

1. What is the phenomenon called when an electric current passes through a conductor and produces heat? [1]
(A) Magnetic effect (B) Chemical effect
(C) Heating effect (D) Luminous effect
2. Three resistors of 2 Ω, 3 Ω, and 6 Ω are connected in parallel. What is their equivalent resistance? [1]
(A) 11 Ohm (B) 6 Ohm
(C) 1 Ohm (D) 0.5 Ohm
3. Which of the following describes electric power? [1]
(A) Total energy (B) Rate of energy
(C) Energy per charge (D) Force on charge
4. If a conductor's length is doubled and its area of cross-section is halved, how does its resistance change? [1]
(A) Halved (B) Doubled
(C) Four times (D) Unchanged
5. What is the SI unit of resistivity? [1]
(A) Ohm (B) Ohm-meter
(C) Siemens (D) Volt/Ampere
6. When resistors are connected in parallel, what remains the same across them? [1]
(A) Current (B) Resistance
(C) Power (D) Potential difference
7. Why is an electric fuse typically made of a material with a low melting point? [1]

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|-------------------------|----------------------|
| (A) Conduct electricity | (B) Melt easily |
| (C) Be strong | (D) Resist corrosion |

8. If the potential difference across a resistor is doubled, what happens to the current flowing through it, assuming resistance is constant? [1]

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|------------------------|-----------------------|
| (A) Current halves | (B) Current doubles |
| (C) Current quadruples | (D) Current unchanged |

9. What is the SI unit of electric current? [1]

- | | |
|------------|-------------|
| (A) Volt | (B) Ohm |
| (C) Ampere | (D) Coulomb |

10. Which principle is utilized in the working of an electric fuse? [1]

- | | |
|---------------------|---------------------|
| (A) Magnetic effect | (B) Chemical effect |
| (C) Heating effect | (D) Luminous effect |

Section B - Very Short Answer Questions

Answer the following questions briefly.

5 × 2 = 10

11. Differentiate between series and parallel combinations of resistors based on how the total resistance changes when more resistors are added. [2]

12. Why is it generally not advisable to connect household appliances in series? [2]

13. State two advantages of connecting electrical appliances in parallel in a domestic circuit. [2]

14. Define electric potential difference and state its SI unit. [2]

15. Consider two identical electrical heaters. If one is connected in series and the other in parallel to the same power supply, which one would draw more current and why? [2]

Section B - Short Answer

Answer in 2-3 sentences.

5 × 3 = 15

16. Derive the expression for the equivalent resistance when three resistors R1, R2, and R3 are connected in series. [3]

17. Describe the working principle of an electric fuse. [3]

18. Discuss the advantages of parallel circuits over series circuits in domestic wiring. [3]

19. Differentiate between series and parallel combinations of resistors based on current flow and potential difference. [3]

20. Derive the expression for the equivalent resistance when two resistors R1 and R2 are connected in series, using Ohm's Law. [3]

Section D - Long Answer Questions

Answer the following questions in detail.

$3 \times 5 = 15$

- 21.** An electric heater is rated 1500 W, 220 V. Calculate the current drawn by the heater and the power consumed if it operates at 200 V. [5]
- 22.** State Joule's Law of heating. On what factors does the heat produced in a resistor depend? [5]
- 23.** Describe the heating effect of electric current and state Joule's Law of heating. Explain two practical applications of the heating effect of electric current. [5]